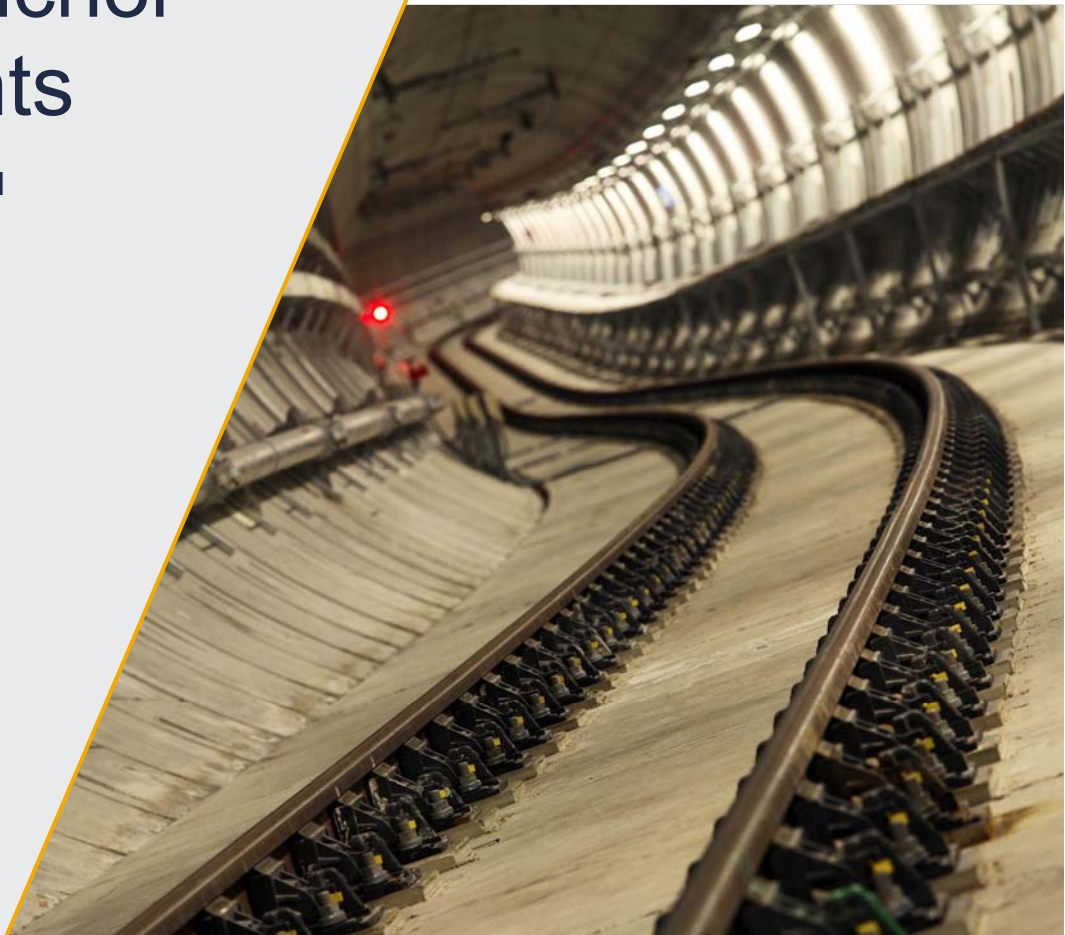


PANDROL

Pandrol Anchor Components

PRODUCT SPECIFICATION



LC-L

29 March 2021

Partners in excellence



Revision History

Version	Date	Authorised by	Comments
5	11 March 2019	Head of Engineering and Support	Presentation format updated and screwspikes included. Illustrations of where to measure coating thickness added. Mod 3354
6	29 March 2021	Head of Engineering and Support	Square headed bolts and screws added and correction to standard and Table 4 for Bolts with Trapezoidal threads ECR15386

Contents

1. Scope	4
2. Hexagon Headed Bolts and Screws	5
3. Masonry and Foundation Bolts (Splayed Anchors)	6
4. Square Headed Bolts and Screws	7
5. Bolts with Trapezoidal Threads	8
6. Double Ended Studs	9
7. Hexagon socket head cap screws	10
8. Screwspikes	11
9. Hexagon plain nuts	12
10. Prevailing Torque Hexagon Nuts with non-metallic insert (Nyloc Nuts)	13
11. Plain washers	14
12. Disc Springs	15
13. Bonded seal washers	16

General note

The manufacturer shall check that specification reference and issue code on this document corresponds exactly to the code on purchase order.

The use of equivalent International Standards to those specified below is subject to written agreement from Pandrol Head Office, Technical Department.

The equipment used during these tests shall be included in a recognised calibration system and shall be traceable to International Standards.

For packaging, labelling and shipping requirements refer to the purchase order.

1. Scope

To specify the requirements for Pandrol brand anchor components.

This document does not include requirements for threaded inserts / dowels.

2. Hexagon Headed Bolts and Screws

Pandrol hexagon headed bolts and screws conform to the requirements of ISO 4014 and Table 1.

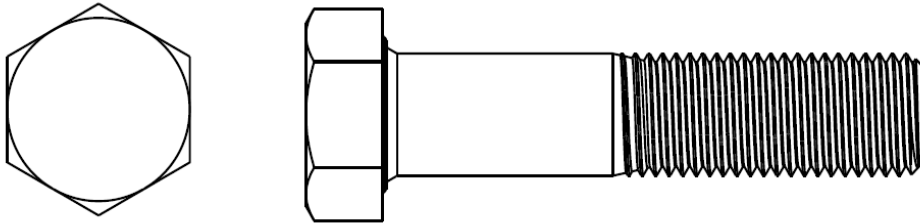


Figure 1 – Hexagon Headed Bolt

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference		
Material		Steel		
Mechanical properties	International Standard	ISO 898-1		
	Property class	8.8	10.9	12.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness		Zinc Flake coating to ISO 10683, 5 to 50µm thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 1 – Product requirements for Hexagon Headed Bolts and Screws

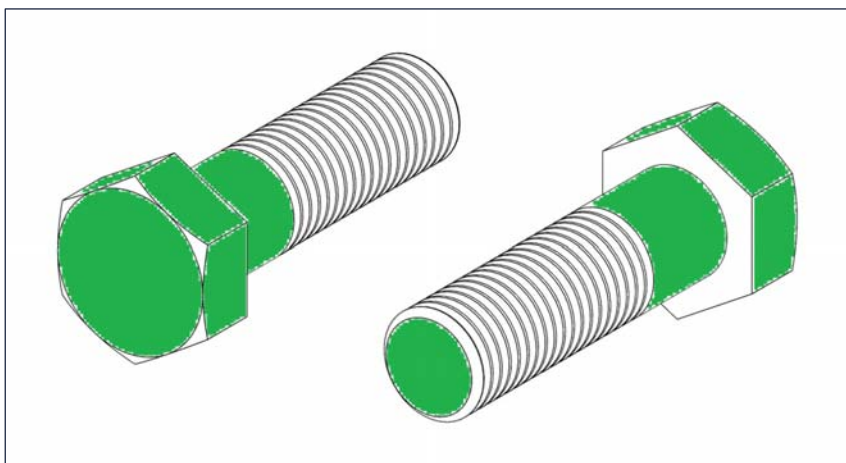


Figure 2 – Showing areas for measuring coating thickness (green) on Hexagon Headed Bolts and Screws

3. Masonry and Foundation Bolts (Splayed Anchors)

Pandrol splayed anchors conform to the requirements of DIN 529:1984, Type D and Table 2.



Figure 3 – Splayed Anchor

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference		
Material		Steel		
Mechanical properties	International Standard	ISO 898-1		
	Property class	8.8	10.9	12.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness		
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 2 – Product requirements for Masonry and Foundation Bolts (Splayed Anchors)

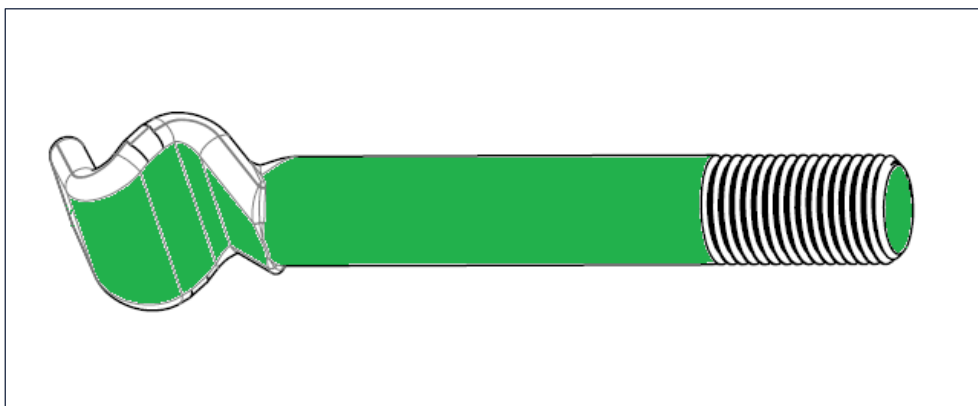


Figure 4 – Showing areas for measuring coating thickness (green) on Splayed Anchors

Square headed bolts and screws added and correction to manufacturing standard for Bolts with Trapezoidal threads

4. Square Headed Bolts and Screws

Pandrol square headed bolts and screws conform to the requirements of BS 64 and Table 3.

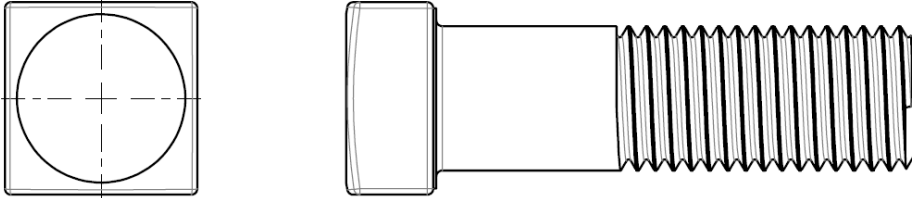


Figure 5 – Square Headed Bolt

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference	
Material		Steel	
Mechanical properties	International Standard	ISO 898-1	
	Property class	8.8	10.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	Zinc Flake coating to ISO 10683, 5 to 50µm thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness	
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
		Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness	

Table 3 – Product requirements for Square Headed Bolts and Screws

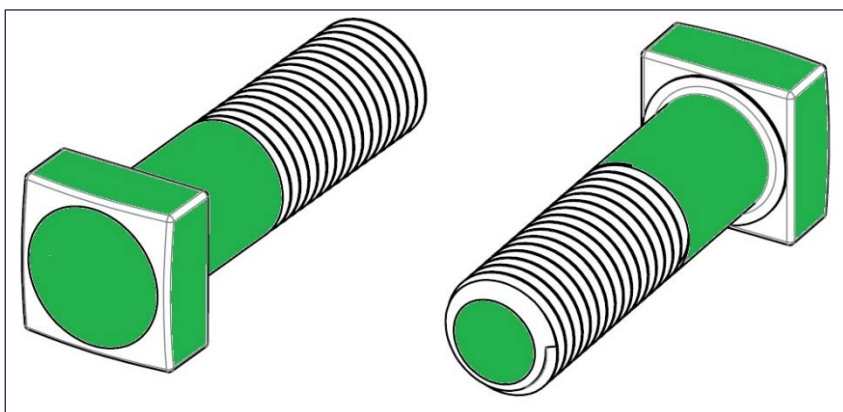


Figure 6 – Showing areas for measuring coating thickness (green) on Square Headed Bolts and Screws

5. Bolts with Trapezoidal Threads

Pandrol bolts with trapezoidal threads conform to the requirements of NF F 50-014 and Table 4.

The head shape for bolts with trapezoidal threads may also be Hexagon Headed or Hexagon Socket Head Cap and reference should be made to the appropriate standards and tables in this document.

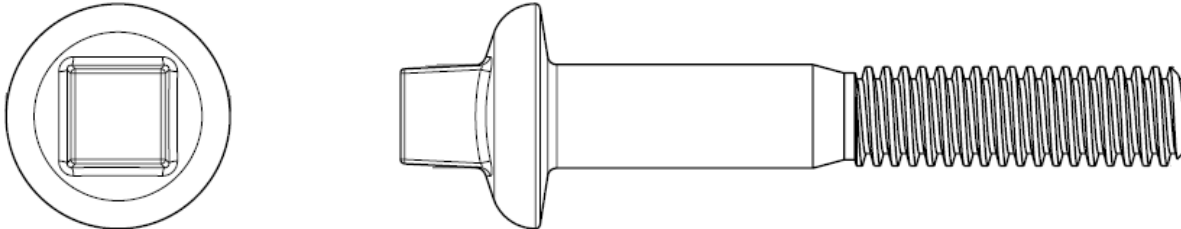


Figure 7 – Bolt with Trapezoidal Thread

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference	
Material		Steel	
Thread	International Standards	ISO 2901, ISO 2902, ISO 2903	
Mechanical properties	International Standard	ISO 898-1	
	Property class	5.6	8.8
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness	
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
		Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness	

Table 4 – Product requirements for Bolts with Trapezoidal Threads

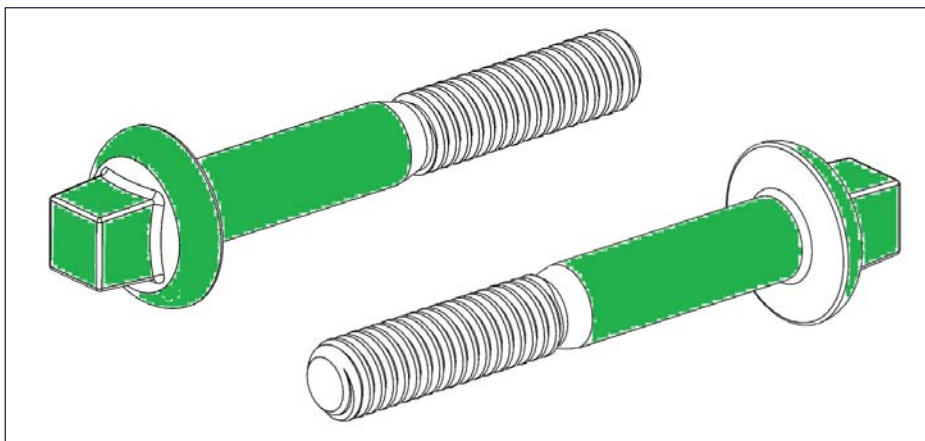


Figure 8 – Showing areas for measuring coating thickness (green) on Bolts with Trapezoidal Threads

6. Double Ended Studs

Pandrol double ended studs conform to the requirements of BS 4439 and Table 5.



Figure 9 – Double Ended Stud

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Item		Specification / reference		
Material		Steel		
Mechanical properties	International Standard	ISO 898-1		
	Property class	8.8	10.9	12.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 5 – Product requirements for Double Ended Studs

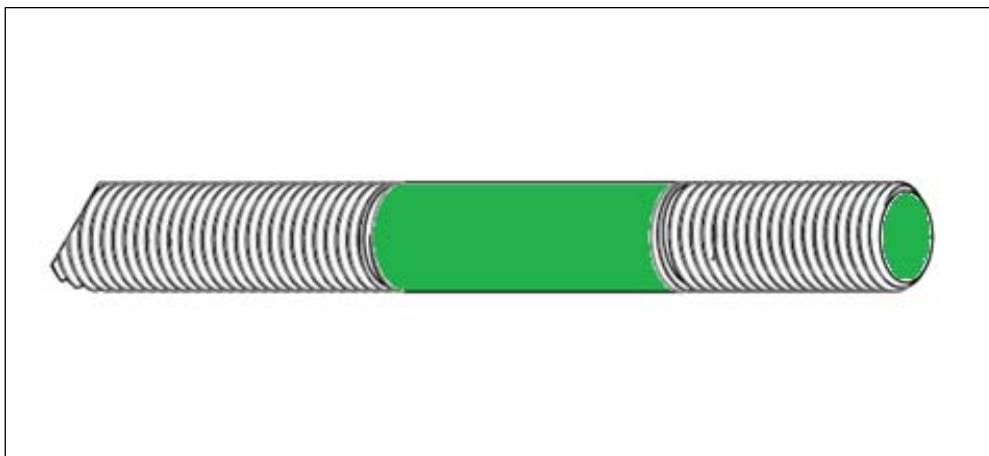


Figure 10 – Showing areas for measuring coating thickness (green) on Double Ended Studs

7. Hexagon socket head cap screws

Pandrol Cap Screws conform to the requirements of ISO 4762 and Table 6.

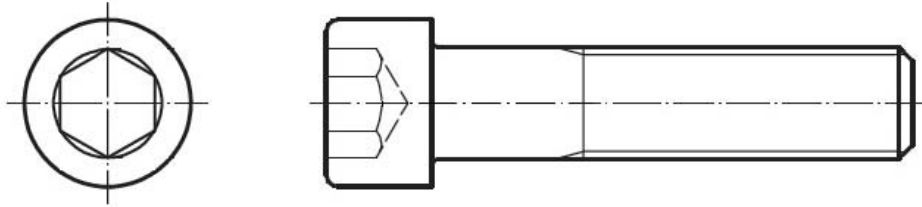


Figure 11 – Cap Screw

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference		
Material		Steel		
Mechanical properties	International Standard	ISO 898-1		
	Property class	8.8	10.9	12.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness		
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 6 – Product requirements for Hexagon socket head cap screws

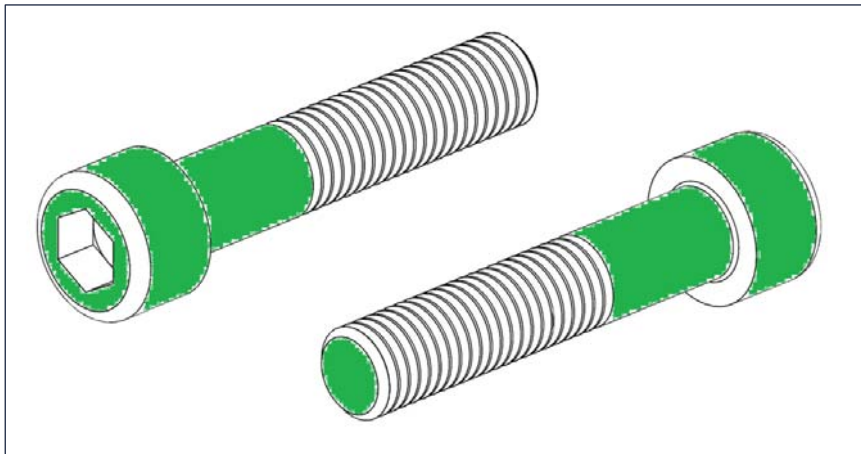


Figure 12 – Showing areas for measuring coating thickness (green) on Cap Screws

8. Screwspikes

Pandrol Screwspikes conform to the requirements of UIC 864-1 and Table 7.

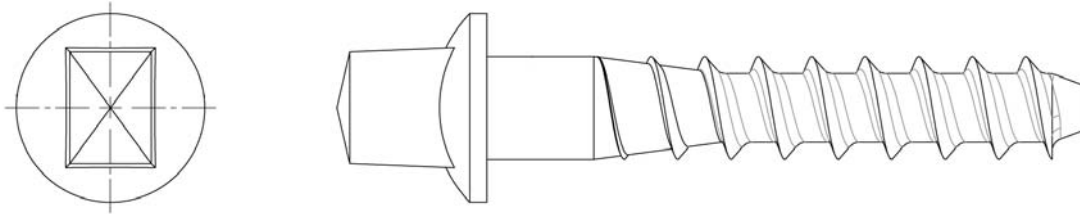


Figure 13 –Screwspike

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference	
Material		Steel	
Mechanical properties	International Standard	ISO 898-1	
	Property class	5.6	8.8
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness	
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
		Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness	

Table 7 – Product requirements for Screwspikes

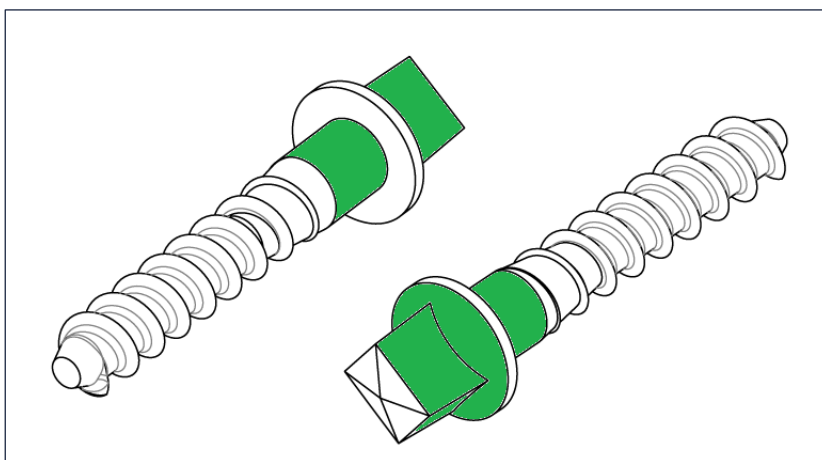


Figure 14 – Showing areas for measuring coating thickness (green) on Screwspikes

9. Hexagon plain nuts

Pandrol hexagon plain nuts conform to the requirements of ISO 4032 and DIN 934 and Table 8.

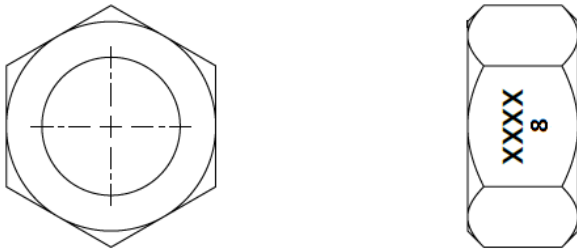


Figure 15 – Plain Nut

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Item		Specification / reference		
Material		Steel		
Thread	Tolerance	6AZ (for clearance of coated male threads)		
	International Standard	ISO 724, ISO 965-1, ISO 965-5		
Mechanical properties	International Standard	ISO 898-2		
	Property class	6	8	10
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness		Zinc Flake coating to ISO 10683, 5 to 50µm thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 8 – Product requirements for Hexagon plain nuts

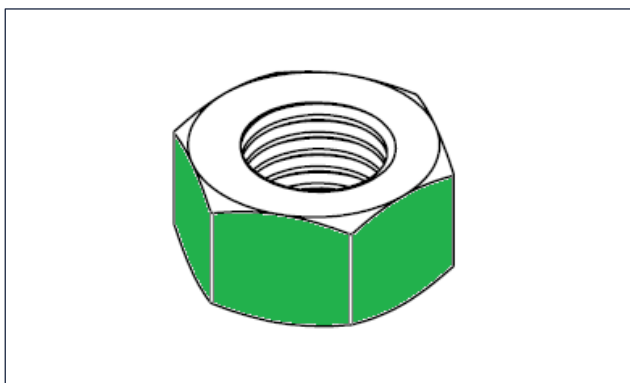


Figure 16 – Showing areas for measuring coating thickness (green) on Plain Nuts

10. Prevailing Torque Hexagon Nuts with non-metallic insert (Nyloc Nuts)

Pandrol Nyloc thin nuts conform to the requirements of ISO 10511 and DIN 985, and Table 9.

Pandrol Nyloc full nuts conform to the requirements of ISO 7040, and Table 9.

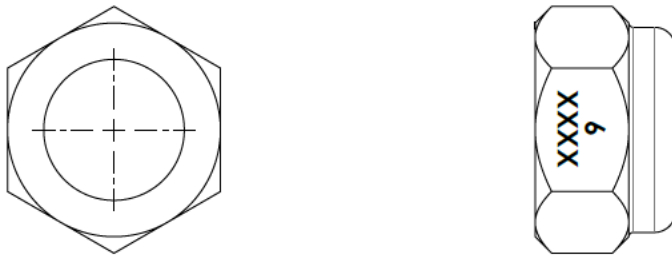


Figure 17 – Nyloc Nut

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Item		Specification / reference		
Material		Steel and Polyamide		
Thread	Tolerance	6AZ (for clearance of coated male threads)		
	International Standard	ISO 724, ISO 965-1, ISO 965-5		
Mechanical and performance properties	International Standard	ISO 898-2, ISO 2320		
	Property class	6	8	10
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness		Zinc Flake coating to ISO 10683, 5 to 50µm thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 9 – Product requirements for Prevailing Torque Hexagon Nuts with non-metallic insert (Nyloc Nuts)

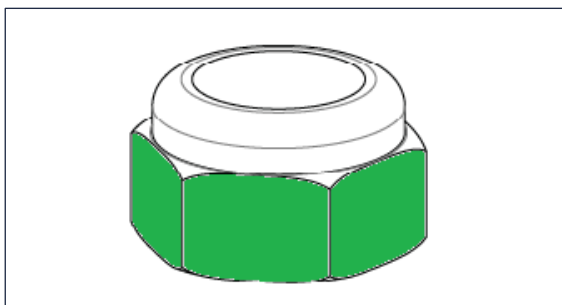


Figure 18 – Showing areas for measuring coating thickness (green) on Nyloc Nuts

11.Plain washers

Pandrol plain washers conform to the requirements of BS 4320 and Table 10.

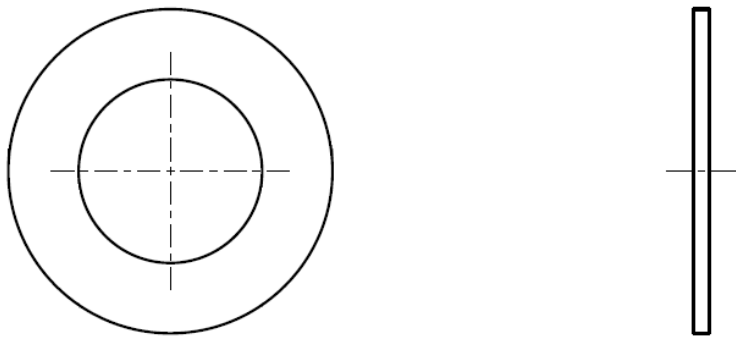


Figure 19 – Plain Washer

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Item		Specification / reference
Material		Steel as specified in BS 4320
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness
		Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness

Table 10 – Product requirements for Plain Washers

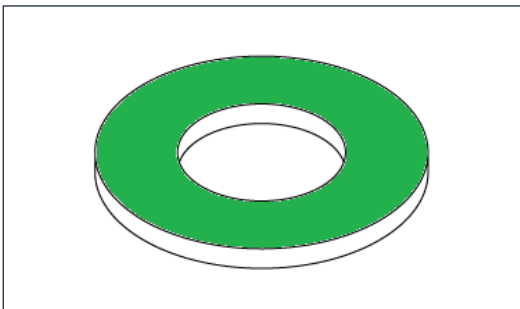


Figure 20 – Showing areas for measuring coating thickness (green) on Plain Washers

12. Disc Springs

Pandrol disc springs conform to the requirements of EN 16983 and Table 11.

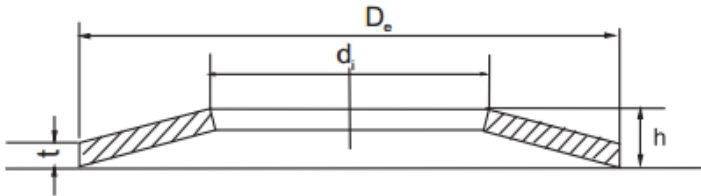


Figure 21 – Disc Spring

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: EN 16983 has replaced DIN 2093

Item		Specification / reference
Material		Spring steel
	Hydrogen embrittlement test	Parallel bearing surface test to ISO 15330
Coating	Corrosion protection	Mechanical zinc plate with iridescent / yellow chromate conversion to ISO 12683, 12µm minimum local thickness

Table 11 – Product requirements for Disc Springs

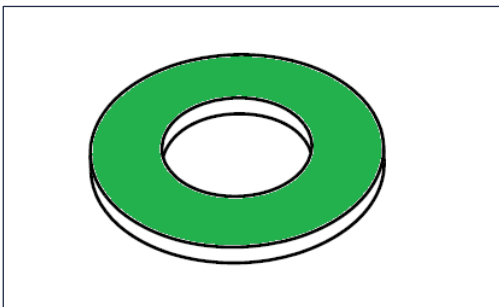


Figure 22 – Showing areas for measuring coating thickness (green) on Disc Springs

13. Bonded seal washers

Pandrol bonded seal washers conform to the requirements of the Pandrol drawing and Table 12.

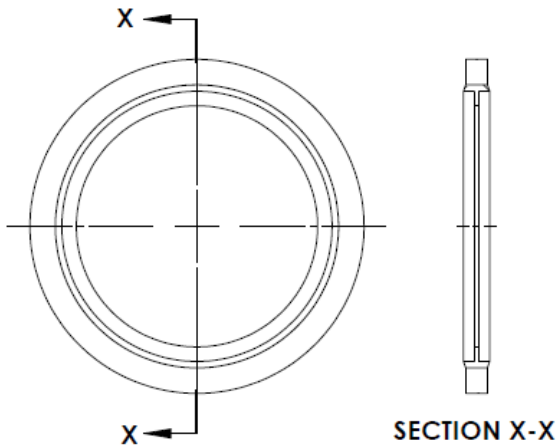


Figure 23 – Bonded Seal Washer

- NOTE: Dimensions and tolerances specified on the relevant Controlled Pandrol drawing shall be used.

Item		Specification / reference
Material		Steel and Nitrile Rubber
Coating	Corrosion protection	Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness
	Alternative corrosion protection	Mechanical zinc plate and iridescent / yellow chromate conversion to ISO 12683, 12µm minimum local thickness

Table 12 – Product requirements for Bonded Seal Washers

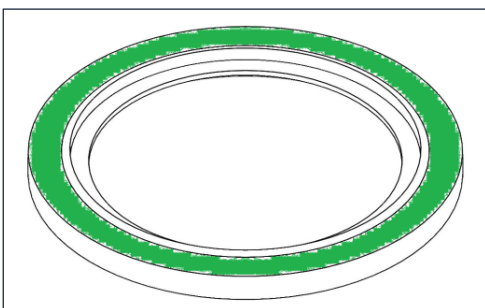


Figure 24 – Showing areas for measuring coating thickness (green) on Bonded Seal Washers

PANDROL

Find out more at

pandrol.com

Pandrol Rail Fastening Systems
Gateford Road,
Worksop,
Nottinghamshire
S81 7AX
UK

+44 (0)1909 476101
www.pandrol.com

© Pandrol

Partners in excellence